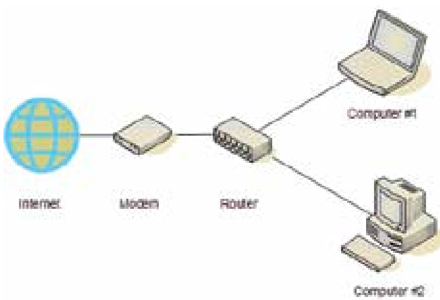
is way more than most users need so any significant reduction in that speed would still be more than enough for the average user.

Types of hardware

A typical home user or small business user can manage their own networks by understanding what kind of hardware they are using and what would be needed to improve upon their current situation. The image below shows a typical network setup found in most home, here the modem and the router are shown separately whereas in most instances they'll be the in one device.

Modem

A modem (Modulator-demodulator) is a device that modulates an analog signal to carry digital information and then demodulates the same at the receiving end to extract the digital information. The purpose was to utilise existing infrastructure namely the telephone network to increase the reach of the internet.



Router

Now the modem creates a network between two computers or two nodes. However, what if the data were to be sent to another place without actually going through the hassle of demodulating and modulating at each and every node, this is where the router comes into the picture. Data is sent over the internet in the form of packets. Each packet has a header which is a label of sorts. It tells where the packet originated from and where it is headed and when it was created etc. Basically it tells the router where it wants to go and the router accordingly dispatches the packet in the right direction. Routers can be wired or wireless. A router may or may not have a modem in it, this is an important thing to remember when buying wireless routers, if you plan on using just one router to connect to the internet then having a router with a modem is needed.

Switch

Routers and switches are very similar in function except for a few differences, routers have very few ports (4-8) while switches can have quite